

AI & Neural Networks

Course Project - Final Report

From Predictive Maintenance to Energy Monitoring:
Six Stages of Deep Learning on Time-Series Data

Course: AI & Neural Networks

University: Astana IT University (AITU)

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Abstract

This report summarises a six-stage course project in which the same team of students went end-to-end from raw time-series to a deployable anomaly-detection module. Stages 1 to 3 were developed on the SCANIA Component X predictive-maintenance dataset, where the goal was regression on a time-to-event target. At Stage 4 the team switched to the Appliance Energy Prediction dataset, converted the problem to binary classification above/below the median consumption, and retained the same chronological 70/15/15 split for the remainder of the project. Across the six stages we implemented and compared MLP, DNN, LSTM, 1D CNN, a denoising autoencoder + CNN pipeline, and a Siamese network trained with a hand-coded contrastive loss. The report collects the key metrics, the architectures, the qualitative observations, and the integrated analysis across all stages.

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1. Project Overview and Datasets

1.1 Goal and methodology

The project follows a ten-week syllabus in which each stage builds on artefacts produced in the previous one. Stages 1 to 3 treat the SCANIA predictive-maintenance dataset; after Stage 3 the team switched to the Appliance Energy Prediction dataset, which provides cleaner and denser time series that are better suited to the architectures required by the later stages (CNN, autoencoder, Siamese). Every stage is implemented in Python with Keras / TensorFlow, pandas, NumPy, scikit-learn and matplotlib, and documented in a dedicated Jupyter notebook plus a stage-level report.

1.2 Dataset A - SCANIA Component X (Stages 1-3)

SCANIA Component X is a real-world multivariate time series collected from 33,000+ heavy-duty trucks. The goal is predictive maintenance: predict component failure from sporadic sensor readouts. Raw data is anonymised (sensor names like 171_0, 666_0, 459_15). We processed `train_operational_readouts.csv` (~1.2 GB), `train_specifications.csv` and `train_tte.csv` into a single frame keyed on `vehicle_id`. Because the raw file is large, the team worked on a 50,000-row sample and used `length_of_study_time_step` as the regression target for Stages 2 and 3 (renaming `in_study_repair` to `class_label` for the classification-oriented EDA).

1.3 Dataset B - Appliance Energy Prediction (Stages 4-6)

`energydata_complete.csv` contains 19,735 observations sampled every 10 minutes over about 4.5 months. Features include indoor temperatures (T1-T9), indoor humidities (RH_1-RH_9), outdoor weather (T_out, Press_mm_hg, RH_out, Windspeed, Visibility, Tdewpoint) and two random reference variables (rv1, rv2). Four cyclical time features are engineered from the timestamp (hour_sin, hour_cos, dow_sin, dow_cos) for 32 inputs in total. A binary target `HighConsumption` is created by thresholding Appliances at its median (60 Wh). The dataset is split chronologically 70/15/15 into train / validation / test, and windowed into samples of shape (24, 32) - four hours of history at 10-minute cadence. `StandardScaler` is fit on training features only to prevent leakage.

2. Stage 1 - EDA and Feature Engineering (SCANIA)

Stage 1 focused on understanding the SCANIA data and producing a clean feature matrix for downstream modelling:

- Load and merge operational readouts, vehicle specifications and time-to-event labels on vehicle_id (sampled to 50,000 rows, ~130 columns).
- Timestamp features: lifecycle_progress (0-1 per vehicle), hour_bin, day_bin, is_weekend, obs_number.
- Missing-value strategy: forward-fill within each vehicle -> backward-fill -> median fill.
- Outliers: IQR method with 1.5 x IQR capping per sensor.
- Engineered features: lag (1, 2, 3 steps) and rolling means (windows 3, 6, 12) for the top 3 correlated sensors, categorical time_of_day, and diff (rate-of-change) features.

Output: processed_data.csv (50,000 rows, 130+ columns) plus seven EDA figures (correlation, distributions, time series, temporal structure, class distribution, missing values, outliers).

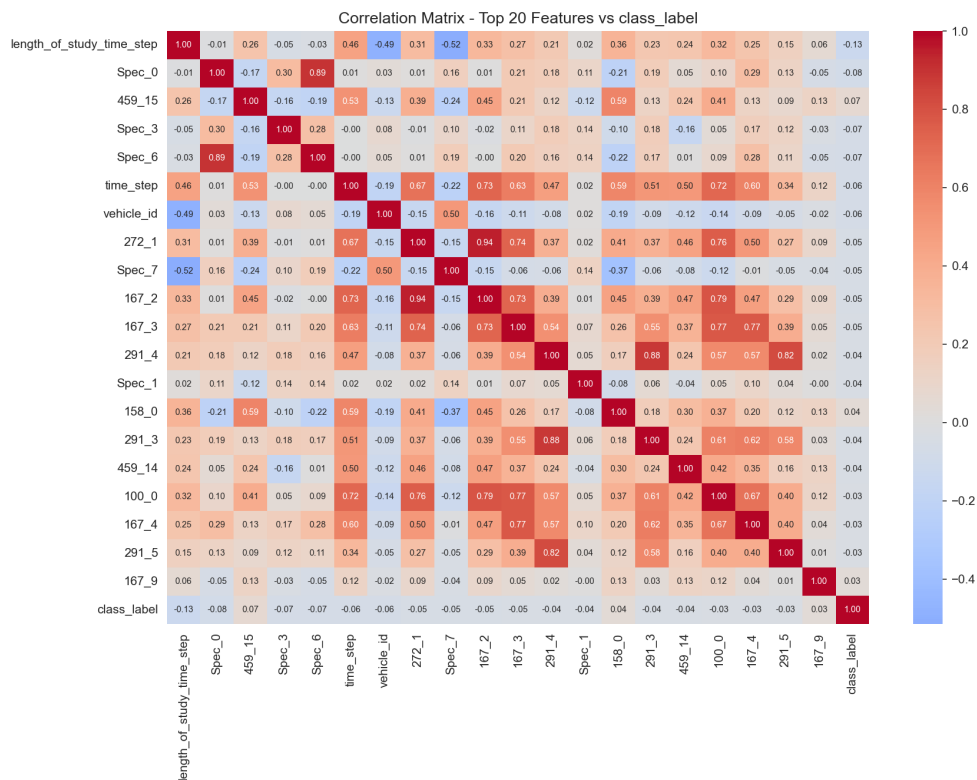


Figure 1: SCANIA feature correlation heatmap (top sensors).

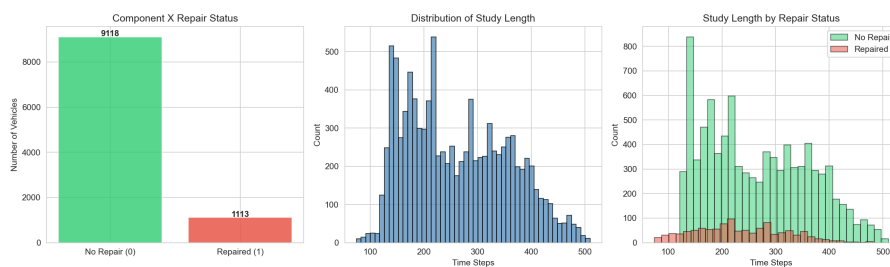


Figure 2: Binary repair-label distribution (class imbalance).

3. Stage 2 - MLP and DNN Regression (SCANIA)

Stage 2 built dense regression baselines on the Stage 1 processed data. The target is `length_of_study_time_step` (operational time to event), treated as a continuous value. Two architectures were trained:

- MLP: a compact two-hidden-layer network (Dense -> ReLU -> Dropout -> Dense -> ReLU -> Dense(1, linear)).
- DNN: a deeper version with four hidden layers, larger widths and slightly stronger regularisation.

Both were trained with Adam, MSE loss, EarlyStopping on validation RMSE.

3.1 Results

Model	Split	RMSE	MAE
MLP	Train	59.35	45.88
MLP	Valid	60.30	49.64
MLP	Test	71.30	58.95
DNN	Train	53.56	40.21
DNN	Valid	50.90	40.72
DNN	Test	67.61	55.18

The DNN generalised better than the MLP (test RMSE 67.61 vs 71.30, test MAE 55.18 vs 58.95). Both models showed the usual train/test gap caused by per-vehicle sequence length variability in the 50k sample.

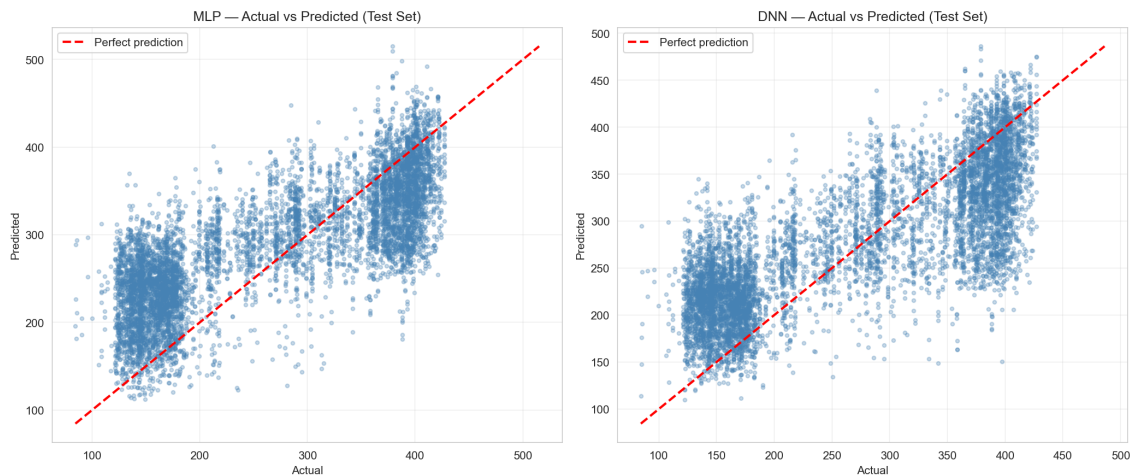


Figure 3: DNN predictions vs actual time-to-event (test set).

4. Stage 3 - LSTM Regression (SCANIA)

Stage 3 introduced recurrence. Variable-length per-vehicle sequences were built with padding and a Masking layer, and a single-layer LSTM followed by Dense output was trained with Adam + MSE and EarlyStopping. A window-length experiment swept the history window from 6 to 48 steps.

4.1 Key results

Model	Test RMSE	Test MAE
MLP (Stage 2)	71.30	58.95
DNN (Stage 2)	67.61	55.18
LSTM window=12 (saved)	59.71	49.92
LSTM window=48 (best observed)	54.04	44.04

The saved model (lstm_best.h5) was selected by validation RMSE and used a 12-step window, improving test RMSE over the Stage 2 DNN by 7.90 and test MAE by 5.25. The lowest test RMSE observed in the window experiment was at window=48 (RMSE 54.04), confirming that longer history helps on this dataset, even though the conservative validation-based selection preferred a shorter window.

Padding + masking allowed a fair comparison across window lengths while preserving per-vehicle boundaries.

Dataset transition - Stage 4 onwards

At Stage 4 the team switched to the Appliance Energy Prediction dataset. Two reasons drove the switch:

- The SCANIA sample had very short per-vehicle sequences (longest ~16 observations), which makes 1D CNN / autoencoder / Siamese architectures - all of which benefit from long, dense windows - awkward to apply.
- The Energy dataset provides a uniform 10-minute sampling grid for ~4.5 months, so rolling 24-step windows with causal Conv1D are natural.

Stages 4, 5 and 6 all share the same chronological 70/15/15 split on this dataset (train 13,790, val 2,936, test 2,937 windows of shape (24, 32)) so their test metrics are directly comparable.

5. Stage 4 - 1D CNN Binary Classification (Energy)

Stage 4 formulated the problem as binary classification: HighConsumption = 1 when Appliances > 60 Wh (median), 0 otherwise, yielding a near-balanced target (45.6% / 54.4%). A compact 1D CNN was trained with a hyperparameter search over two configurations:

Input(24, 32) -> Conv1D(filters, kernel, causal, ReLU) -> MaxPool1D(2) -> Flatten -> Dense(64, ReLU) -> Dropout(0.3) -> Dense(1, Sigmoid)

5.1 Results

filters	kernel	Accuracy	Precision	Recall	F1
64	3	0.8229	0.8539	0.7557	0.8018
128	5	0.7960	0.8015	0.7572	0.7787

Config A (filters=64, kernel=3) was selected. On the test set it scored Accuracy 0.7995, Precision 0.7687, Recall 0.8498, F1 0.8072, ROC-AUC 0.8851. A logistic regression baseline on the flattened 24x32 windows (768 features) was competitive (F1 0.8398), showing that a large share of the class signal is captured by linear combinations of scaled features - especially the cyclical time features.

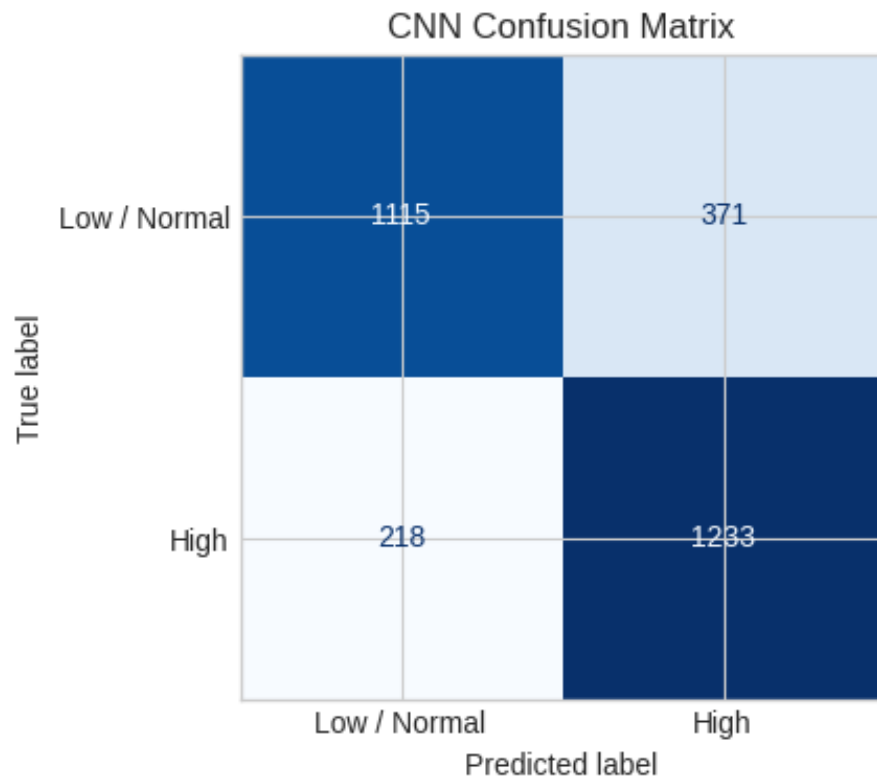


Figure 4: Stage 4 CNN confusion matrix on the test set.

6. Stage 5 - Denoising Autoencoder (Energy)

Stage 5 evaluated the noise-robustness of the Stage 4 classifier and trained a dense autoencoder to recover the clean signal from noise-corrupted inputs.

6.1 Noise simulation and autoencoder

Zero-mean Gaussian noise with $\sigma = 0.5$ (in standardised units) was added to the test features to simulate sensor malfunction. A fully-connected autoencoder was trained on clean data only with MSE loss:

Encoder: Input(24 x 32) -> Dense(512) -> Dense(256) -> Dense(128 bottleneck)

Decoder: Dense(256) -> Dense(512) -> Dense(24 x 32, linear)

6.2 Reconstruction quality

Variant	MSE vs clean	Noise reduction
AE on clean (baseline)	0.0970	-
Noisy (before AE)	0.2504	-
Denoised (after AE)	0.1230	50.9%

6.3 Downstream classifier impact

Variant	Accuracy	Precision	Recall	F1
Clean	0.7995	0.7687	0.8498	0.8072
Noisy	0.7869	0.7748	0.8015	0.7879
Denoised	0.7692	0.7620	0.7746	0.7683

Noise alone costs the CNN about 1-2 p.p. of accuracy - the Conv1D receptive field already averages over local perturbations. The denoising autoencoder halves reconstruction MSE, but running the classifier on its smooth output actually loses another ~1 p.p. of accuracy. Pixel-level reconstruction and downstream classification are not the same objective: the AE's smoothing discards high-frequency detail that the CNN relies on. This trade-off is a classic finding and motivates joint or task-aware denoising training.

Signal Reconstruction: Clean vs Noisy vs Denoised

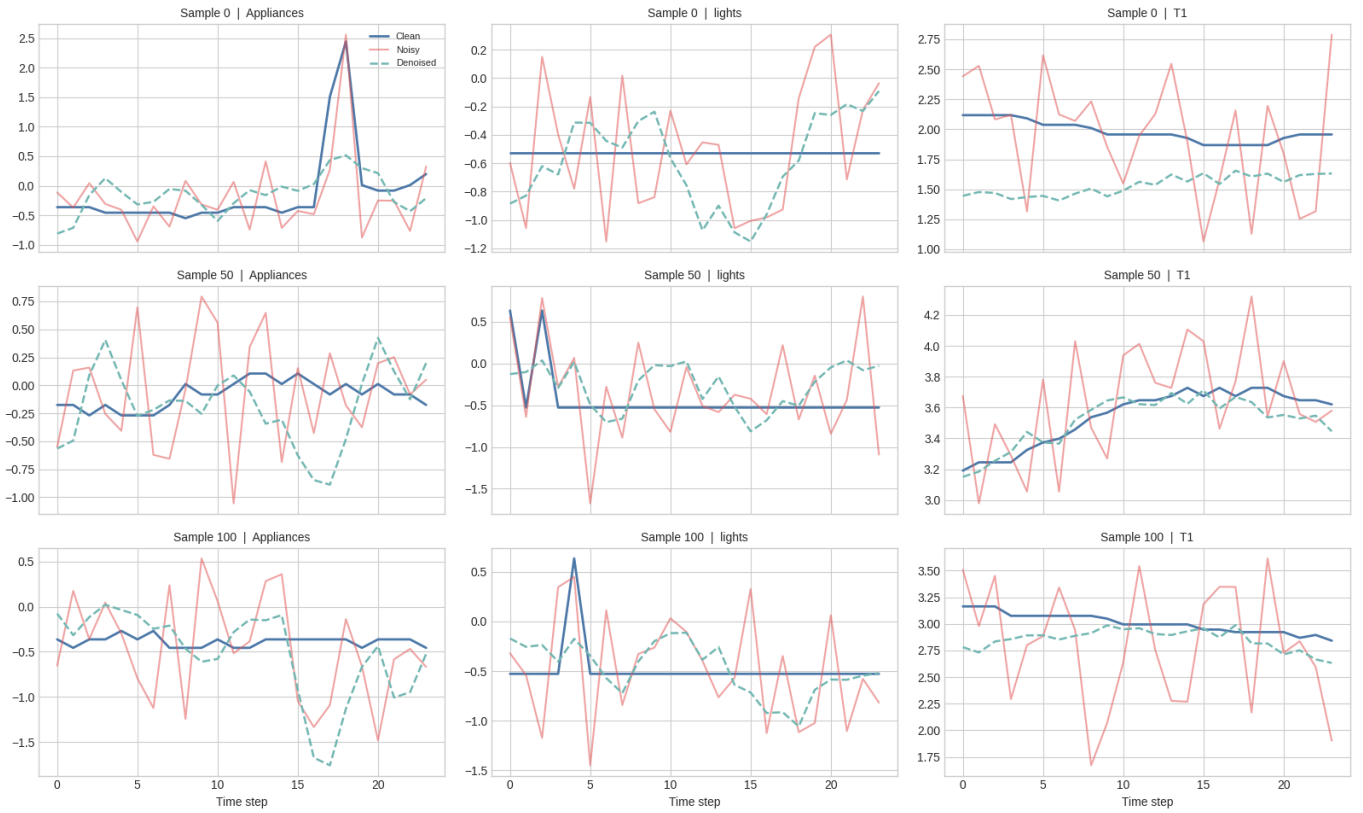


Figure 5: Clean vs noisy vs denoised signal for sample windows / channels.

7. Stage 6 - Siamese Network for Anomaly Detection (Energy)

Stage 6 leaves behind supervised labels at inference time and asks: how far is this window from anything we have ever seen as normal? A Siamese network is trained with a hand-coded contrastive loss on balanced pairs.

7.1 Pair construction

- Positive ($Y = 0$, similar): two windows with HighConsumption = 0.
- Negative ($Y = 1$, dissimilar): one normal + one high-consumption window.

8,000 pairs per class for training (16,000 total) and 2,000 per class for validation / test. The Y convention matches Hadsell, Chopra and LeCun (2006).

7.2 Architecture and loss

Shared subnetwork (reuses the Stage 4 backbone plus a second conv block + projection):

Input(24, 32) -> Conv1D(64, k=3, causal) -> MaxPool1D(2) -> Conv1D(128, k=3, causal) -> MaxPool1D(2) -> Flatten -> Dense(128, ReLU) -> Dropout(0.3) -> Dense(128) -> L2Normalise (128-d embedding)

$$D(a, b) = \sqrt{\sum_i (a_i - b_i)^2 + \epsilon}$$

$$L(Y, D) = (1 - Y) * 0.5 * D^2 + Y * 0.5 * \max(0, m - D)^2 \quad (m = 1.0)$$

7.3 Pair-level evaluation

Validation threshold sweep selected $t = 0.2833$ (val pair accuracy 0.7395). On test pairs:

Pair metric	Accuracy	Precision	Recall	F1
Siamese	0.7452	0.7667	0.7050	0.7346

7.4 Anomaly detection on individual windows

A reference bank of 2,000 normal embeddings is built from the training set. For each test window the anomaly score is the minimum Euclidean distance to any reference embedding. Test ROC-AUC against HighConsumption reaches 0.8685. Thresholded at Youden's J on the ROC curve: Accuracy 0.786, Precision 0.830, Recall 0.712, F1 0.767.

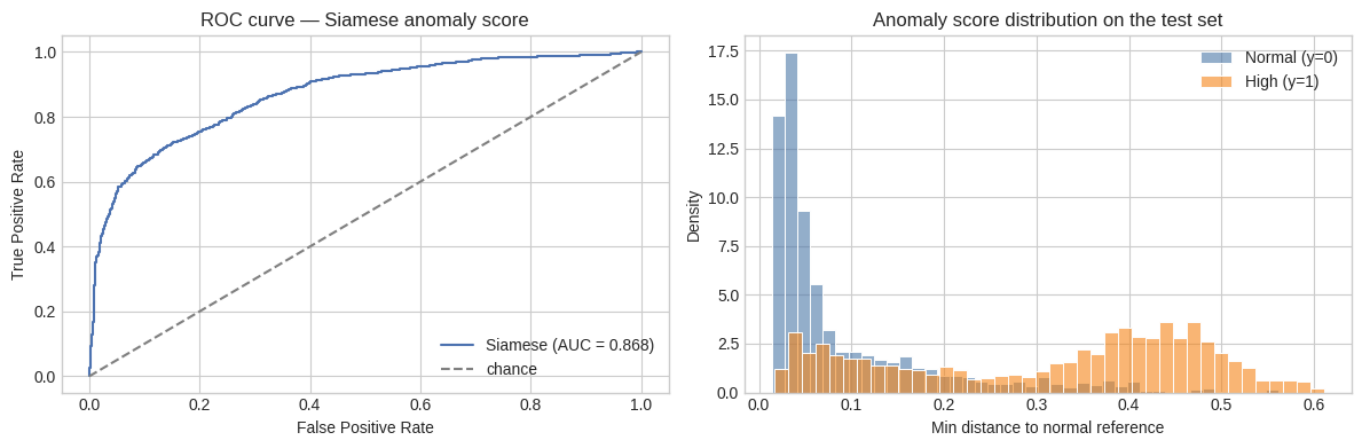


Figure 6: Stage 6 Siamese ROC curve and anomaly-score distribution.

8. Integrated Comparison

8.1 SCANIA stages (regression)

Model	Test RMSE	Test MAE
MLP (Stage 2)	71.30	58.95
DNN (Stage 2)	67.61	55.18
LSTM window=12 saved (Stage 3)	59.71	49.92
LSTM window=48 best (Stage 3)	54.04	44.04

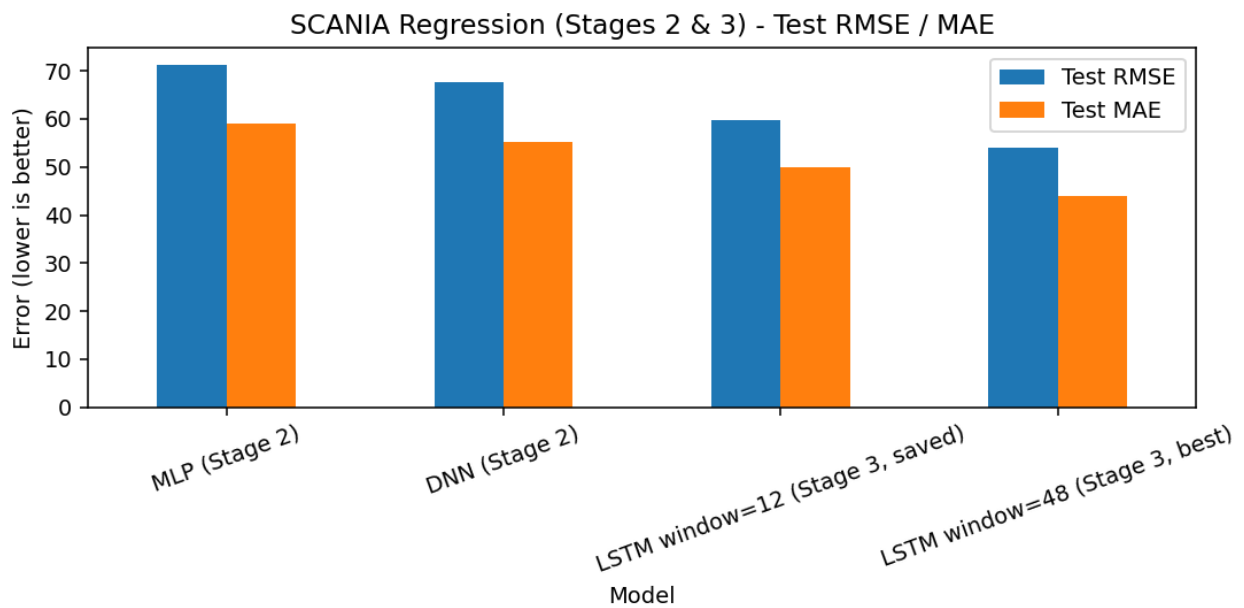


Figure 7: SCANIA regression test error across Stage 2 / 3 models.

8.2 Energy stages (classification and anomaly detection)

Model	Acc	Prec	Rec	F1	AUC
CNN classifier (Stage 4)	0.7995	0.7687	0.8498	0.8072	0.8851
CNN on noisy test (Stage 5)	0.7869	0.7748	0.8015	0.7879	-
CNN on AE-denoised test (Stage 5)	0.7692	0.7620	0.7746	0.7683	-
Siamese + contrastive (Stage 6)	0.7858	0.8304	0.7119	0.7666	0.8685

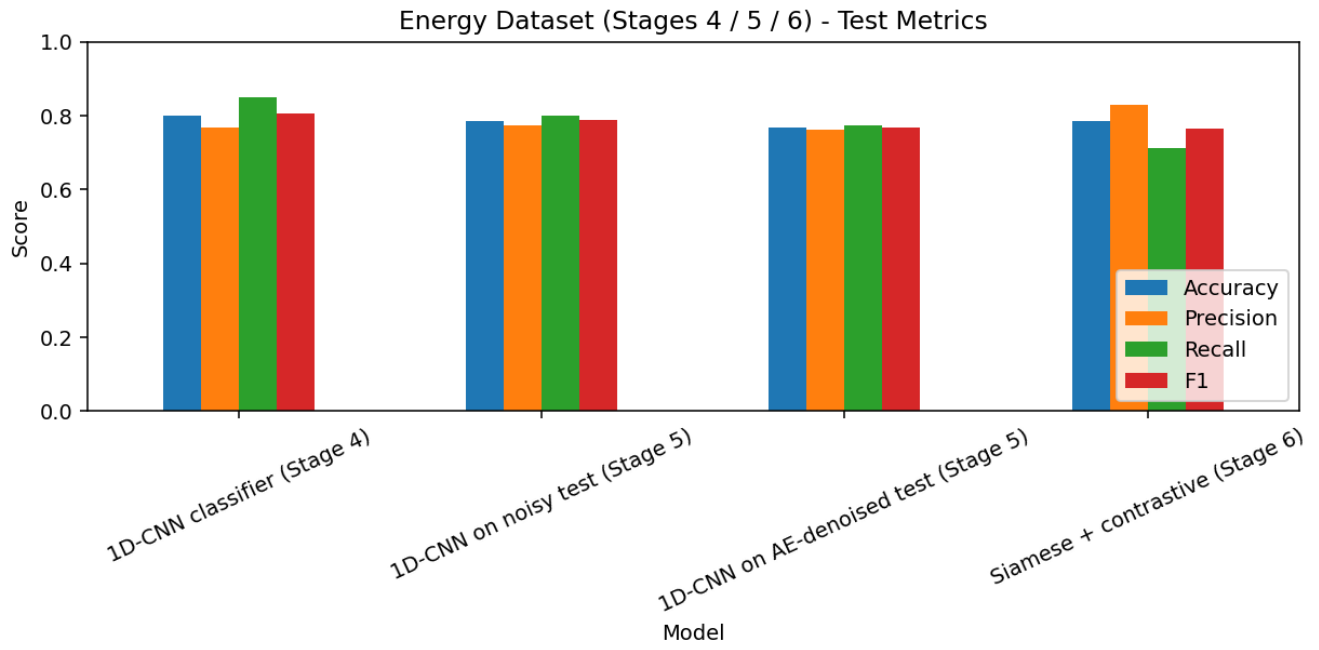


Figure 8: Stage 4 / 5 / 6 models on the energy test split.

9. Discussion

9.1 Architecture vs task

- Forecasting / regression (SCANIA): adding recurrence via an LSTM gave a clear win over dense MLP / DNN baselines (test RMSE 59.7 vs 67.6 at the saved window, with 54.0 achievable at longer history). Temporal structure matters, and per-vehicle masking makes it tractable.
- Supervised classification (Energy, Stage 4): a compact 1D CNN with causal convolutions is the strongest closed-set classifier on clean inputs (F1 0.807, ROC-AUC 0.885). A logistic-regression baseline on flattened windows is surprisingly competitive (F1 0.84), a reminder that for tabular-style short windows linear models with cyclical time features capture a large share of the signal.
- Noise robustness and denoising (Stage 5): a fully-connected denoising AE halves reconstruction MSE under $\sigma = 0.5$ noise, but its smooth output slightly hurts the downstream CNN. Reconstruction fidelity and downstream discriminability are not the same objective.
- Open-set anomaly detection (Stage 6): a Siamese network trained with contrastive loss reaches ROC-AUC 0.868 - on par with the supervised CNN (0.885) - using only a pool of 2,000 normal references at inference time. It offers higher precision (0.830) at the cost of lower recall (0.712) than the supervised CNN.

9.2 Engineering lessons

- Chronological splits and fitting preprocessing only on train-time data were critical on both datasets to avoid leakage.
- Causal padding kept Conv1D honest about temporal order in Stages 4 and 6.
- Reusing the Stage 4 backbone inside the Siamese subnetwork reduced the number of moving pieces and made the two approaches directly comparable.
- Pair-construction convention ($Y = 0$ similar vs $Y = 1$ dissimilar) has to match the contrastive-loss formula signs - the team followed Hadsell, Chopra and LeCun (2006) to stay consistent with the textbook formulation.
- The dataset switch at Stage 4 improved our ability to compare architectures fairly, at the cost of breaking apples-to-apples comparability with Stages 2 and 3.

10. Conclusions

The six-stage project delivered a complete deep-learning workflow over time-series data, covering EDA, dense and recurrent regression, convolutional classification, denoising via autoencoders, and metric-learning for anomaly detection. On the SCANIA dataset, an LSTM with per-vehicle masking outperformed dense baselines on regression. After the dataset switch, on the Appliance Energy data the final Siamese anomaly detector matched the supervised Stage 4 CNN's ROC-AUC (0.868 vs 0.885) without using any labels at inference time, using only a small bank of normal reference windows. The three energy-stage models target different questions - closed-set classification, noise-robust reconstruction, and open-set anomaly scoring - and together form a complementary monitoring toolkit. All artefacts (notebooks, saved models, CSVs, figures, per-stage reports) are committed under the Assignment1 / Assignment2 / Assignment3 / Assignment4 / Assignment5 / Assignment6 directories in the project repository.

References

- Hadsell, R., Chopra, S., & LeCun, Y. (2006). Dimensionality reduction by learning an invariant mapping. CVPR.
- Candanedo, L. M., Feldheim, V., & Deramaix, D. (2017). Data driven prediction models of energy use of appliances in a low-energy house. Energy and Buildings.
- SCANIA CV AB. SCANIA Component X predictive-maintenance dataset (course-provided extract).